MICHELE MAIULLARI

Aeronautical & Mechanical Engineer

CONTACT

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DETAILS

Date of birth: 27.03.2001
Immediate availability for international travel & relocation
EU driving licence – category B
Italian citizen · valid passport

SOFTWARE & TOOLS

CAD / PLM
CATIA · 3DEXPERIENCE · Autodesk Inventor · AutoCAD · FreeCAD

FEA & Simulation
Ansys (structural, fatigue, loads) · MATLAB · Simulink

MBSE
CATIA Magic System Architect (SysML)

Programming
Python · Fortran · Arduino

AI / Machine Learning
LLMs · open-source models · performance & weight evaluation

Dev & OS
Git / GitHub · Linux · Windows · macOS

Office
Excel · Word · PowerPoint

LANGUAGES

- Italian — Native
- English — Proficient
- German — Basic (in course)
- French — Basic (in course)

SOFT SKILLS

- Cross-functional, multidisciplinary coordination, teamwork in international environemnt
- Stakeholder management
- Technical & requirements documentation
- Organization & time management

CERTIFICATIONS

- Leonardo Aerotech Academy
- AMTECH – Additive Manufacturing Technologies
- Training Course for Workers

PROFILE

Aeronautical (MSc) and mechanical (BSc) engineer with industry experience at Leonardo Aerostructures and SKF, hands-on across the full development cycle of UAS components: 3D CAD design, load estimation, structural validation, manufacturing, assembly and testing. Applies MBSE, requirements management and V&V from concept to qualification at RWTH Aachen; developing ML-based aerodynamic design tools validated against CFD

CORE SKILLS

- **Structural Analysis:** FEA/FEM · fatigue & load evaluation · root-cause analysis
- **Aircraft Design & Aerodynamics:** flight mechanics · flight simulation · CFD · UAS/UAV design · airworthiness
- **Systems Engineering:** MBSE, V-Model & SysML · requirements management · trade studies · V&V
- **Manufacturing & Quality:** additive manufacturing · rapid prototyping · series production & quality control

EXPERIENCE

Engineering Intern — Aerotech Academy
Oct 2025 – Jul 2026

Leonardo S.p.A · Aerostructures Business Unit — Grottaglie, Italy

Aerospace · Flight Mechanics · Structures · Advanced Materials · Sustainable Manufacturing · Digital Systems · Avionic System · Airworthiness · Certification · Homologation

- Conceptual to detail design of a wing and a quick-release wing–fuselage coupling for MALE-ISR UAS platforms (CATIA V5) with the Aerostructures R&D department.
- Estimated steady-flight aerodynamic and inertial loads and derived design load cases for the structural substantiation of the wing and its fuselage junction.
- Performed static FEA and Safe-Life fatigue assessment (Ansys), verifying margin of safety and supporting sizing iterations loops and contributed to loads and stress documentation used to demonstrate compliance.
- Supported prototype manufacturing, assembly and testing phases.

Additive Manufacturing Training — AMTECH
Mar 2025

IEC Tehnopolis — Nikšić, Montenegro

AM Applications · Business Models · Advanced Industry Seminar by Meltio

- Applied AM theory to real production processes: reverse engineering and prototype development via advanced 3D scanning and printing technologies at TechLab.

Engineering Intern — Quality & Vibration Analysis
Mar – Jul 2023

SKF S.p.A — Modugno, Italy

- Performed quality control on single-row deep-groove ball bearings for electric-automotive applications, classifying rejects by noise frequency band (NVH Analysis).
- Performed vibrational testing with FFT and envelope spectral analysis on EV ball bearings, correlating defect signature to production root causes and corrective action.

EDUCATION

Politecnico di Bari - MSc Aeronautical Engineering
Nov 2023 – Oct 2026

Coursework: Fluid Dynamics · Energy Systems · Aeronautical Materials & Manufacturing · Design & Simulation

- **Thesis (with Leonardo S.p.A):** *"Design and Testing of Lightweight Aeronautical Systems, UAVs and UASs, Built Using Rapid Assembly and Deployment Techniques."*

RWTH Aachen University - Erasmus Exchange
Mar – Oct 2025

- Non-linear FEM · fatigue of structures · fatigue design of lightweight structures · model-based product development.

Politecnico di Bari - BSc Mechanical Engineering
Oct 2020 – Nov 2023

Coursework: Applied & Structural Mechanics · Thermo-Fluids & Energy Systems · Manufacturing & Materials · Design · Mechanical & Thermal Measurements

- **Thesis (with SKF S.p.A):** *"Vibrational Analysis of Single-Row Deep-Groove Ball Bearings for Electric Automotive Applications."*

International Exchange — Italian High School Terra Santa
Sep – Oct 2019

- Haifa, Israel · Immersive academic and cultural exchange

SIDE PROJECTS

AI-Based Airfoil Generation Platform — Project Lead
Jan 2026 – Present

- Leading the development of an AI-based aeronautical software system that automatically generates, optimizes and validates airfoils and control surfaces; driving business development and technical validation with key industry stakeholders.